Module – 5

Assignment Submissions – embedUR Networking Course

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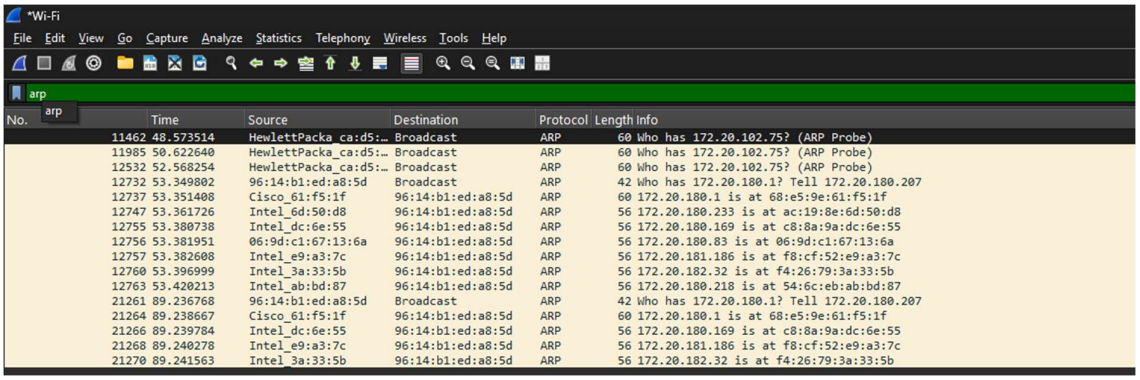
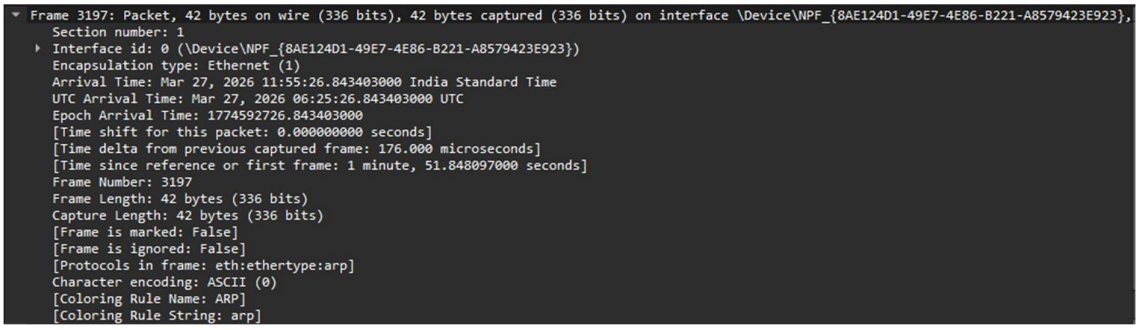
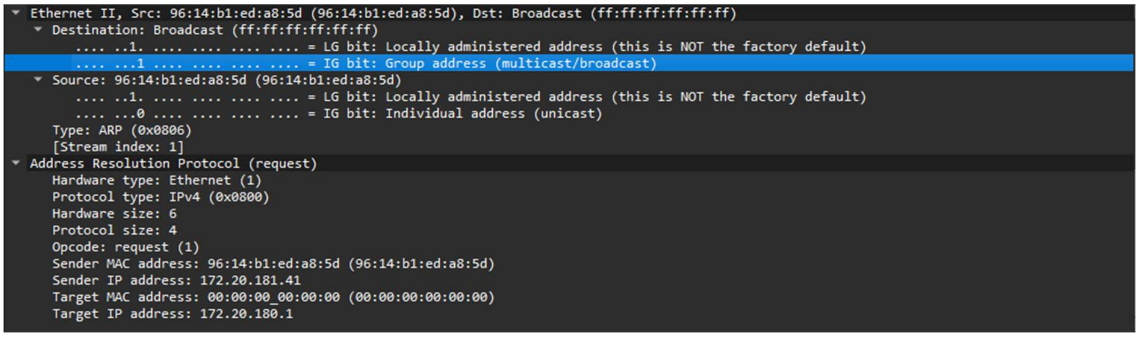
Q1) **Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames, and discuss the role of the sender's IP and MAC address in these packets.**



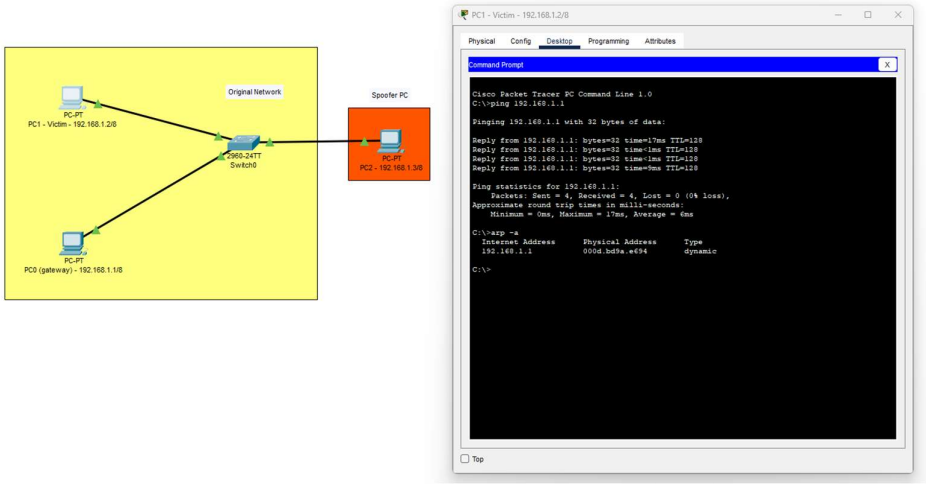




In an ARP packet, the sender's IP address identifies the device that is making the request at the logical (network) level, while the sender's MAC address specifies the actual hardware address where the reply should be delivered. When a device sends an ARP request, it includes both these details so that the receiving device knows who is asking and where to send the response. The IP address provides identification, and the MAC address ensures correct physical delivery of the reply within the local network, making both fields essential for successful communication.

Q2) **Using Packet Tracer, simulate an ARP spoofing attack. Analyze the behavior of devices on the network when they receive a malicious ARP response.**

MAC table via PC1 before spoofing



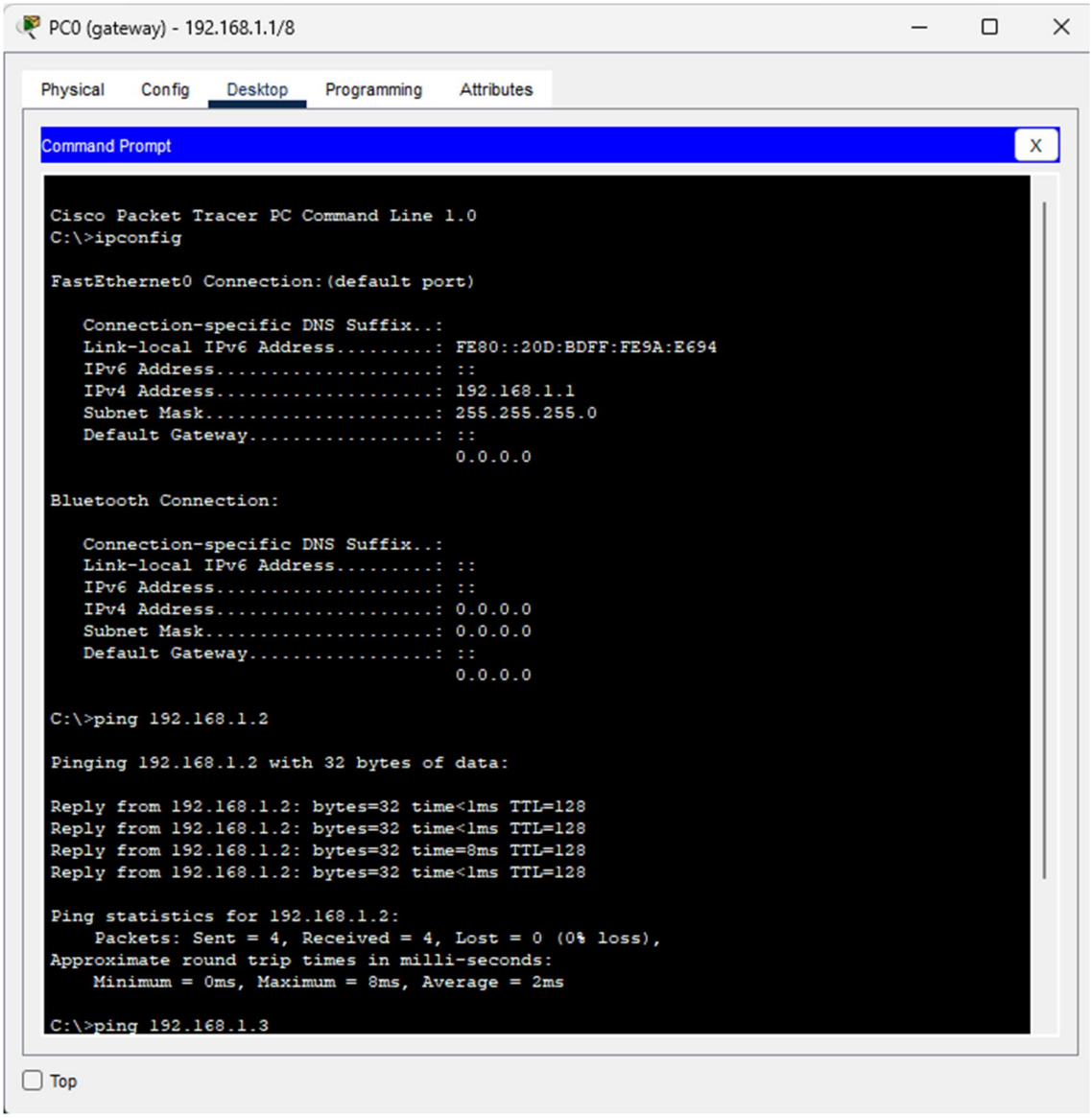
Gateway's MAC address was changed to that of PC2's MAC address

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0001.437b.ab20        dynamic
```

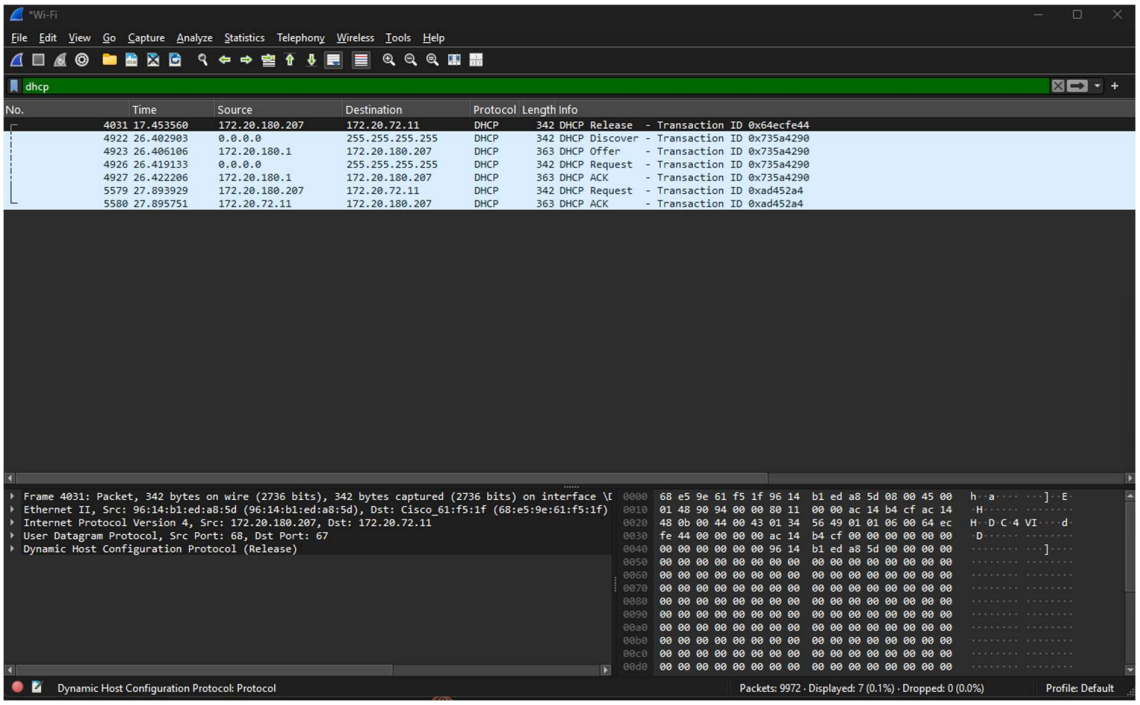
When devices receive a malicious ARP response, they **blindly trust it and update their ARP table** with the new IP–MAC mapping, even if it is fake. In your case, the victim PC replaces the gateway's real MAC address with the attacker's MAC, thinking it is legitimate.

As a result, all traffic meant for the gateway is redirected to the attacker's device. The attacker can then **intercept, modify, or forward the packets** (man-in-the-middle) without the user noticing, which is why the ping still works normally. This behavior highlights a key weakness of ARP—it has **no authentication mechanism**, so devices accept spoofed responses and become vulnerable to attacks like traffic sniffing and session hijacking.

Q3) **Manually configure static IPs on the client devices (like PC or your mobile phone) and verify connectivity using ping.**



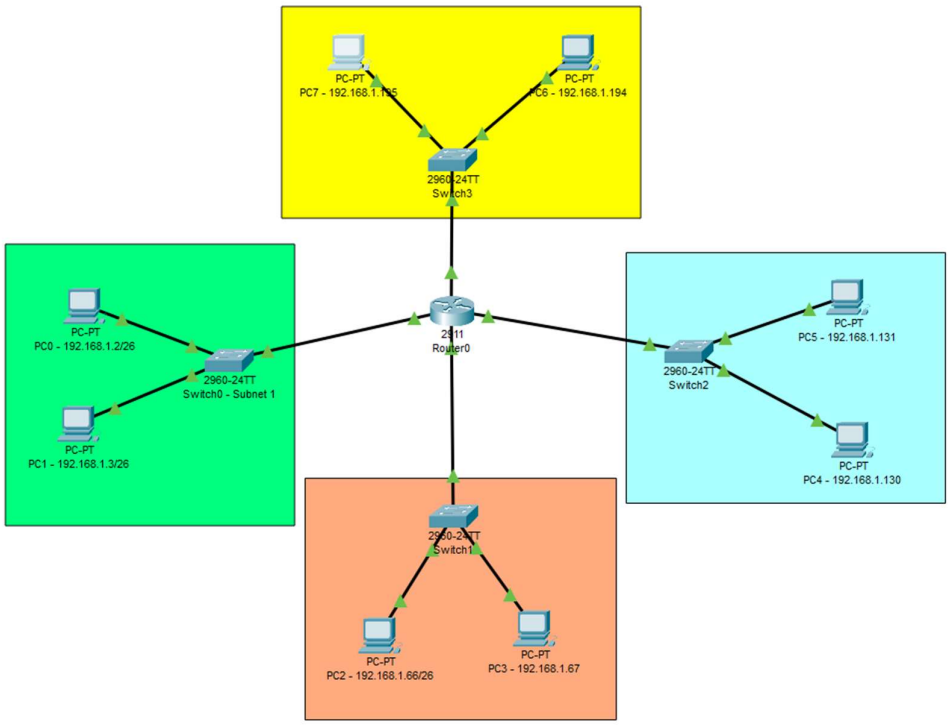
Q4) **Use Wireshark to capture DHCP Discover, Offer, Request, and Acknowledge messages and explain the process.**



DHCP works using a four-step process called **DORA (Discover, Offer, Request, Acknowledge)** to automatically assign IP addresses to devices. First, the client sends a **Discover** message as a broadcast to find available DHCP servers. Then, the server replies with an **Offer**, suggesting an IP address along with configuration details like subnet mask and gateway. The client responds with a **Request**, indicating that it accepts the offered IP address. Finally, the server sends an **Acknowledge (ACK)** message to confirm the assignment, after which the client can start using the IP address for network communication.

Q5) **Given an IP address range of 192.168.1.0/24, divide the network into 4 subnets. Manually calculate the new subnet mask and the range of valid IP addresses for each subnet.**

Assign IP addresses from these subnets to devices in Cisco Packet Tracer and verify connectivity using ping between them.



The network 192.168.1.0/24 can be divided into 4 subnets by borrowing 2 bits,

Resulting in a new subnet mask of 255.255.255.192 (/26).

Each subnet has 64 IP addresses.

The four subnets are

192.168.1.0/26 (usable range 192.168.1.1–192.168.1.62),

192.168.1.64/26 (192.168.1.65–192.168.1.126),

192.168.1.128/26 (192.168.1.129–192.168.1.190),

192.168.1.192/26 (192.168.1.193–192.168.1.254),

with corresponding broadcast addresses at the end of each range.

Q6) **You are given three IP addresses: 10.1.1.1, 172.16.5.10, and 192.168.1.5.**

Task: Identify the class of each IP address (Class A, B, or C). What is the default subnet mask for each class?

Provide the range of IP addresses for each class.

10.1.1.1 – is Class A

Default subnet is 255.0.0.0

Range – 0.0.0.0 – 127.255.255.255

172.16.5.10 – is Class B

Default subnet is 255.255.0.0

Range – 128.0.0.0 – 192.255.255.255

192.168.1.5 – is Class C

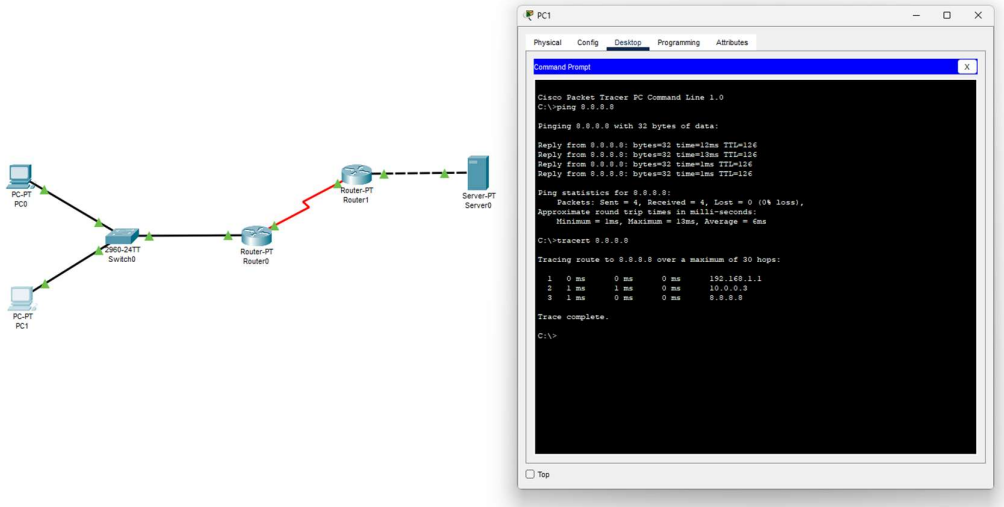
Default subnet is 255.255.255.0

Range – 192.0.0.0 – 223.255.255.255

Q7) **In Cisco Packet Tracer, create a small network with multiple devices (e.g., 2 PCs and a router). Use private IP addresses (e.g., 192.168.1.x) on the PCs and configure the router to perform NAT to allow the PCs to access the internet.**

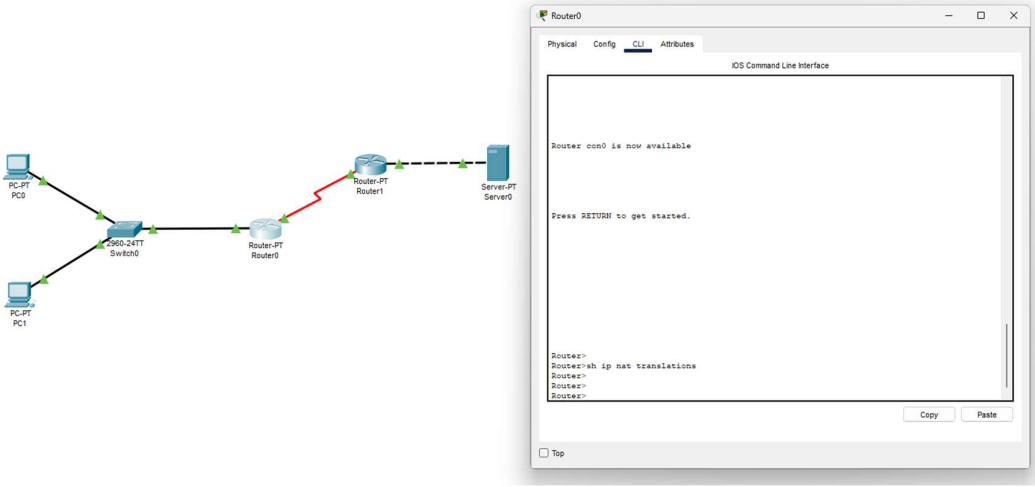
Task: Test the NAT configuration by pinging an external IP address from the PCs and capture the traffic using Wireshark.

What is the source IP address before and after NAT?



Standard dynamic routing

Before NAT




```
Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip nat inside source static 192.168.1.2 10.0.0.2
```

```
Router(config)#

Router(config)#interface FastEthernet0/0

Router(config-if)#ip nat inside

Router(config-if)#ex

Router(config)#

Router(config)#interface Serial2/0

Router(config-if)#ip nat outside

Router(config-if)#ex

Router(config)#

Router(config)#do wr

Building configuration...

[OK]

Router(config)#ex

Router#
```

```
Router#
Router#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  10.0.0.2          192.168.1.2        ---                ---
```

After NAT and Pinging from PC0 to simulated server (8.8.8.8)

```
Router#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.0.0.2:18       192.168.1.2:18     8.8.8.8:18         8.8.8.8:18
icmp 10.0.0.2:19       192.168.1.2:19     8.8.8.8:19         8.8.8.8:19
icmp 10.0.0.2:20       192.168.1.2:20     8.8.8.8:20         8.8.8.8:20
icmp 10.0.0.2:21       192.168.1.2:21     8.8.8.8:21         8.8.8.8:21
---  10.0.0.2          192.168.1.2        ---                ---
```

Before NAT, the source IP address is **192.168.1.2**, which is the private IP of the PC inside the local network. After NAT, the source IP address is translated to **10.0.0.2**, which is the public (inside global) IP address used by the router to communicate with external networks.